CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 11: Algebraic Expressions

Topic-wise short notes with solved examples

Terms, coefficients and types of algebraic expressions, polynomials and their degree, and the four operations on expressions, including division by a monomial and by a binomial.

1 Terms and expressions

- A **constant** has a fixed value. A **variable** (literal) is a letter that can take different values.
- A **term** is a number, a variable, or a product or quotient of numbers and variables: 5, x, $-3xy$, $\frac{2a}{b}$.
- An **algebraic expression** is one or more terms joined by $+$ or $-$ signs.
- **Coefficient**: any factor of a term is the coefficient of the rest. In $-7x^2y$ the numerical coefficient is -7 and the coefficient of y is $-7x^2$.
- **Like terms** have exactly the same variables with the same powers: $3x^2y$ and $-5x^2y$. Otherwise they are **unlike**: $3x^2y$ and $3xy^2$.

Name	Number of terms	Example
Monomial	1	$4xy$
Binomial	2	$3a - 2b$
Trinomial	3	$x^2 + 5x - 6$
Multinomial	More than 3	$a + b - c + 2d$

2 Polynomials

- A **polynomial** is an expression in which every variable has a **whole-number** power. $x^2 + 3x - 4$ is a polynomial; $x + \frac{1}{x}$ and $\sqrt{x} + 2$ are not.
- **Degree in one variable**: the highest power of the variable.
- **Degree in several variables**: the highest **sum of the powers** in any term.

Example 2.1 Find the degree of $5x^3 - 7x^5 + 2x - 1$ and of $3x^2y^3 - 4xy^2 + 7x^4$.

SOLUTION

Highest power: 5. Sums of powers in the second: 5, 3, 4.

5 and 5

Example 2.2 Write the numerical coefficient of $-\frac{3}{4}a^2bc$ and the coefficient of a^2 in it.

SOLUTION

$-\frac{3}{4}$ and $-\frac{3}{4}bc$

3 Addition and subtraction

- Only **like terms** can be added or subtracted: add or subtract their coefficients.
- **To subtract**, change the sign of every term of the expression being subtracted, then add.
- Removing brackets: a $+$ before a bracket leaves the signs unchanged; a $-$ before a bracket changes every sign.

Example 3.1 Add $3x^2 - 5xy + 2y^2$, $-x^2 + 7xy - 4y^2$ and $4x^2 - 2xy$.

SOLUTION

x^2 terms: $3 - 1 + 4 = 6$. xy terms: $-5 + 7 - 2 = 0$. y^2 terms: $2 - 4 = -2$.

$6x^2 - 2y^2$

Example 3.2 Subtract $4a^2 - 3ab + b^2$ from $7a^2 + 2ab - 5b^2$.

SOLUTION

$7a^2 + 2ab - 5b^2 - 4a^2 + 3ab - b^2$

$3a^2 + 5ab - 6b^2$

Example 3.3 What must be added to $x^2 - 3x + 5$ to get $3x^2 + x - 2$?

SOLUTION

$(3x^2 + x - 2) - (x^2 - 3x + 5)$

$2x^2 + 4x - 7$

4 Multiplication

- **Monomial** $\times$ **monomial**: multiply the coefficients and add the powers of like variables.
- **Monomial** $\times$ **polynomial**: multiply each term by the monomial.
- **Polynomial** $\times$ **polynomial**: multiply each term of one by each term of the other, then combine like terms.
- Signs: like signs give $+$; unlike signs give $-$.

Example 4.1 Multiply $-4a^2b^3$ by $3ab^2c$.

SOLUTION

$-12a^3b^5c$

Example 4.2 Multiply $3x(2x^2 - 5x + 4)$.

SOLUTION

$6x^3 - 15x^2 + 12x$

Example 4.3 Multiply $(2x + 3)(3x - 5)$.

SOLUTION

$6x^2 - 10x + 9x - 15$

$6x^2 - x - 15$

Example 4.4 Multiply $(x - 2)(x^2 + 2x + 4)$.

SOLUTION

$x^3 + 2x^2 + 4x - 2x^2 - 4x - 8$

$x^3 - 8$

5 Division

- **Monomial ÷ monomial**: divide the coefficients and subtract the powers. $\frac{a^m}{a^n} = a^{m-n}$.
- **Polynomial ÷ monomial**: divide each term separately.
- **Polynomial ÷ polynomial (long division)**:
- Arrange both in descending powers. Leave gaps (or write $0x^n$) for missing powers.
- Divide the first term of the dividend by the first term of the divisor to get a term of the quotient.
- Multiply the divisor by that term and subtract. Bring down the next term. Repeat until the remainder has a lower degree than the divisor.
- **Check**: Dividend = Divisor × Quotient + Remainder.

Example 5.1 Divide $18x^4y^3$ by $-6x^2y$.

SOLUTION

$-3x^2y^2$

Example 5.2 Divide $12a^3 - 8a^2 + 4a$ by $4a$.

SOLUTION

$3a^2 - 2a + 1$

Example 5.3 Divide $x^2 + 7x + 12$ by $x + 3$.

SOLUTION

$x^2 \div x = x$; subtract $x^2 + 3x$ to get $4x + 12$. $4x \div x = 4$; subtract $4x + 12$ to get 0.

Quotient $x + 4$, remainder 0

Example 5.4 Divide $2x^3 - 5x^2 + 4x - 3$ by $x - 2$.

SOLUTION

$2x^2$: subtract $2x^3 - 4x^2$, leaving $-x^2 + 4x$. $-x$: subtract $-x^2 + 2x$, leaving $2x - 3$. $+2$: subtract $2x - 4$, leaving 1.

Quotient $2x^2 - x + 2$, remainder 1

6 Simplifying with brackets

- Remove brackets from the **innermost** outward: bar, round ( ), curly { }, square [ ].
- Follow the order of operations within each bracket.

Example 6.1 Simplify $5a - [3a - \{2a - (a - b)\}]$.

SOLUTION

$5a - [3a - \{2a - a + b\}] = 5a - [3a - a - b] = 5a - 2a + b$

$3a + b$

Example 6.2 Find the value of $2x^2 - 3xy + y^2$ when $x = 2$ and $y = -1$.

SOLUTION

$8 + 6 + 1$

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7 Common mistakes

- Adding unlike terms: $3x + 2y$ is not $5xy$, and $x^2 + x$ is not x^3.
- Changing the sign of only the first term when subtracting an expression.
- Adding powers in an addition: $x^2 + x^2 = 2x^2$, not x^4.
- Multiplying powers in a product: $x^2 \times x^3 = x^5$, not x^6.
- Not leaving a gap for a missing power in long division.
- Taking the degree of $3x^2y^3$ as 3 instead of 5.